

Nidhi Mathihalli

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EDUCATION

Massachusetts Institute of Technology

Bachelors of Engineering in Computer Science and Math

Cambridge, MA

August 2023 - May 2027

- o **Concentration:** AI and Machine Learning
- o **Relevant Courses:**
 - o CS: Intro to Machine Learning, Design & Analysis of Algorithms, Deep Learning, NLP, Data Structures
 - o Math: Linear Algebra, Discrete Math, Probability and Random Variables (USAMO qualifier)

WORK EXPERIENCE

BNY (Bank of New York Mellon)

Pittsburgh, PA

Software Engineer Intern — Gen AI

June 2025 – Aug 2025

- Developed an AI SRE Agent and an MCP server to process ServiceNow Incident Tickets, automatically generate code fixes, create test cases, and trigger production deployment, leveraging similar issues & solutions from past 6 months
- Built an AI meeting assistant for Microsoft Teams to capture notes and automatically update ServiceNow Tickets
- Presented work to CTO & senior leadership + Engineering Townhall for impact on productivity & workflow automation
- *Tools Used:* MCP Architecture, OpenAI GPT-4o, Microsoft Graph and Teams APIs, ServiceNow APIs, OAuth2

Martini.ai

Palo Alto, CA

Software Engineer Intern — Fintech

May 2024 – Aug 2024

- Developed financial market signals using graph embeddings on bonds data, nearest neighbors, employee count, rank, and company age to predict Zero-Volatility Spreads for assessing company default risk
- Improved z-spread prediction accuracy, strengthening default rate forecasts and credit ratings for startups
- Increased Spearman coefficient for all companies by +23.7%; for companies with most data, increased by +10.5%
- *Tools Used:* Pytorch Big Graph, GGVec, ProNE, k-Nearest Neighbors, PostgreSQL, Python

RESEARCH ASSISTANT (MIT Computer Science and AI Lab, MIT Lincoln Labs)

3D X-Ray Reconstruction derived from Ultrasound Imaging

Jan 2025 – Present

- Developed an end-to-end pipeline for generating 3D Point Clouds from ultrasound images of arms for tumor detection
- Leveraged Spatial Temporal Attention Heads with CNNs to improve training and validation accuracy by 20%
- *Tools Used:* 3D Gaussian Splatting, CNN, Multi-Attention Head

DreamSat: 3D Model for Novel View Synthesis of Space Objects

Nov 2023 – May 2024

- Created 3D point cloud prototypes from single-image input using DreamGaussian 3D reconstruction and a modified Zero123XL for spacecraft reconstruction; increased PSNR (+2.53%) and SSIM (+2.38%) on 30 images
- *Tools Used:* 3D Gaussian Splatting, DreamGaussian, Zero-shot Image-to-3D via Zero123XL, Git, Stable Diffusion

COURSES & PROJECT WORK

Coursemate (HackMIT): Created AI agent that personalizes course curriculum via past class structures a student succeeded in

OS Project (Fall 2025): Design a multi-processor xv6 operating system for RISC-V with VM, networking, kernels

Computer Architecture (Fall 2024): Developed a RISC-V processor in Minispec; implemented cache, parallel processing, etc.

Currency Detector Project: Developed a transfer learning model on Raspberry Pi to predict currency denominations in INR and USD, which I presented at WSCG 2022 and validated through testing with 3 blind schools

AWARDS & RECOGNITION

Math: USA Math Olympiad Qualifier (Top 0.1%; 2022); 4x American Invitational Math Exam Qualifier (2018-2022)

Science: 1st Place, National Junior Science and Humanities Symposium (2022); 4x 1st Place, ISEF, CA (2019-2023)

Accepted as *first author* to the 2024 IAC International Conference; Paper/Code: <https://spacenvs.github.io/>

SKILLS

Programming: Python, C, C++, C#, JavaScript, React.js, TypeScript, Node.js, Go, SQL, Java

AI Tools: Keras, Tensorflow, PyTorch, OpenAI GPT-4o, Numpy, Pandas, Stable Diffusion

Technologies: Docker, MongoDB, GitHub CoPilot, Figma, Linux, REST APIs, Flask

Other: Statistics, Algorithms, Academic Writing, Databases, Problem Solving, Quantitative Analysis